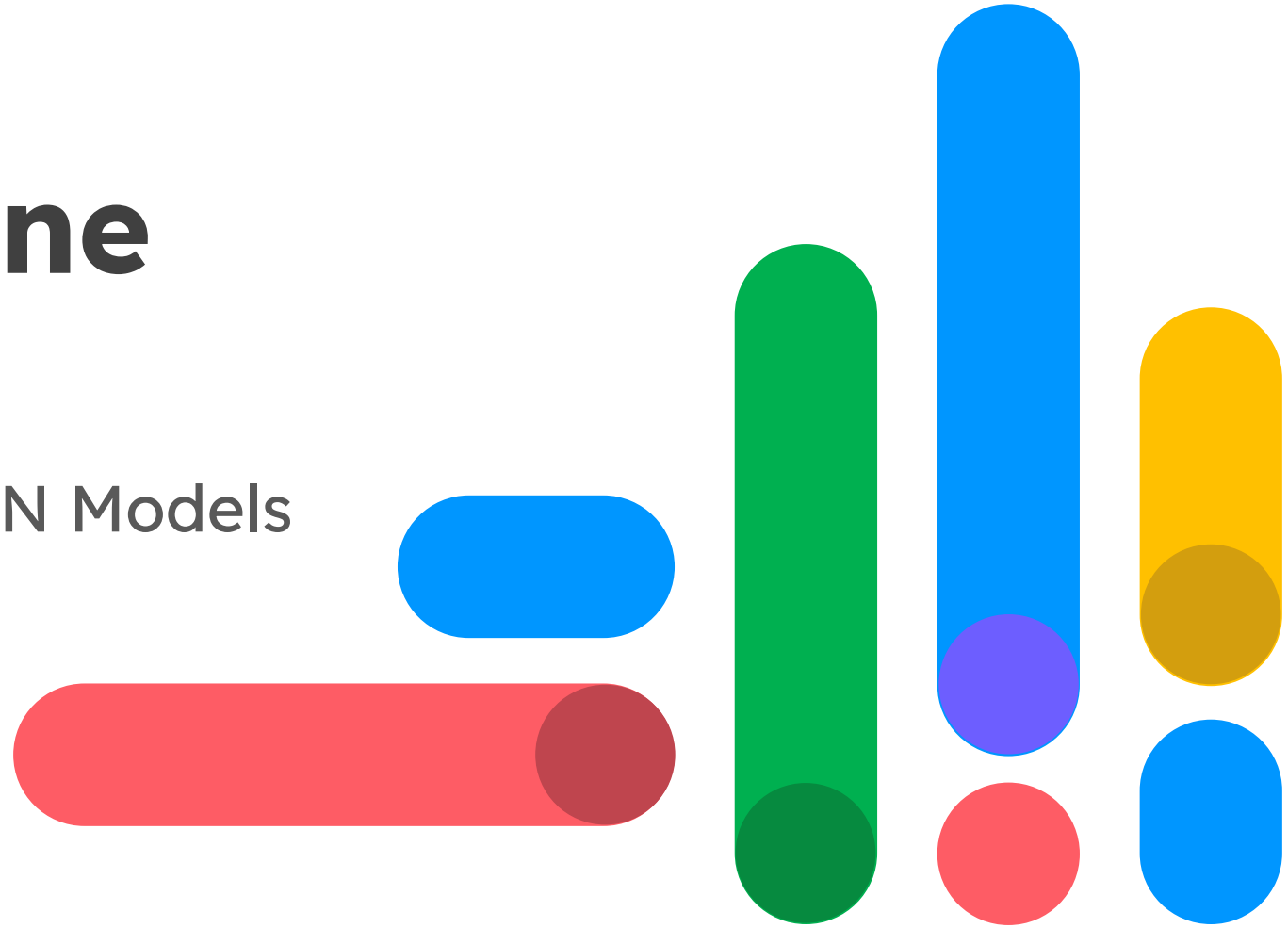
Deep Learning

Applied Machine Learning

Computer Visions :: Basic CNN Models

Assistant Professor | **Chun-Hsiang Chan**

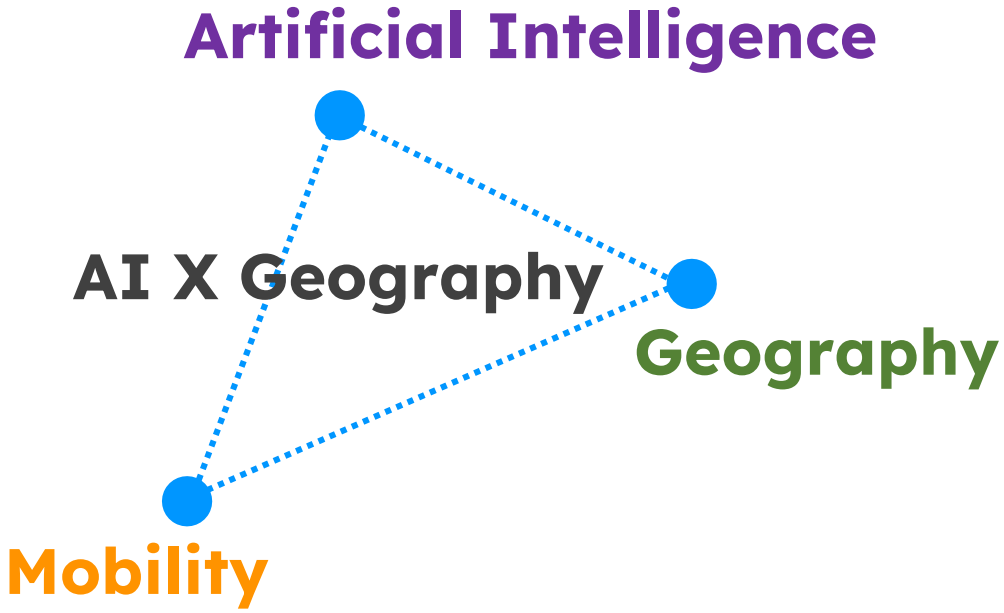
Department of Geography
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Outline

#GeoAI #Mobility #Geography #STDS

01 LeNet
02 AlexNet
03 VGG
04 ResNet
05 Data Augmentation
06 References



LeNet

LeNet is one of the earliest and typical CNN model. This model is used to recognize the document and hand-writing.

Its core idea is to **automatically learn features from images and use those features to identify digits.** For example, earlier layers can learn strokes and edges, while later layers combine this information to recognize shapes.

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278-2324.

LeNet

Convolutional layers use **weight sharing**, allowing the same feature detector to operate at different positions in an image while reducing the number of parameters. Subsampling reduces feature map size and sensitivity to small shifts in position. During training, the model adjusts its parameters through backpropagation.

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278-2324.

LeNet

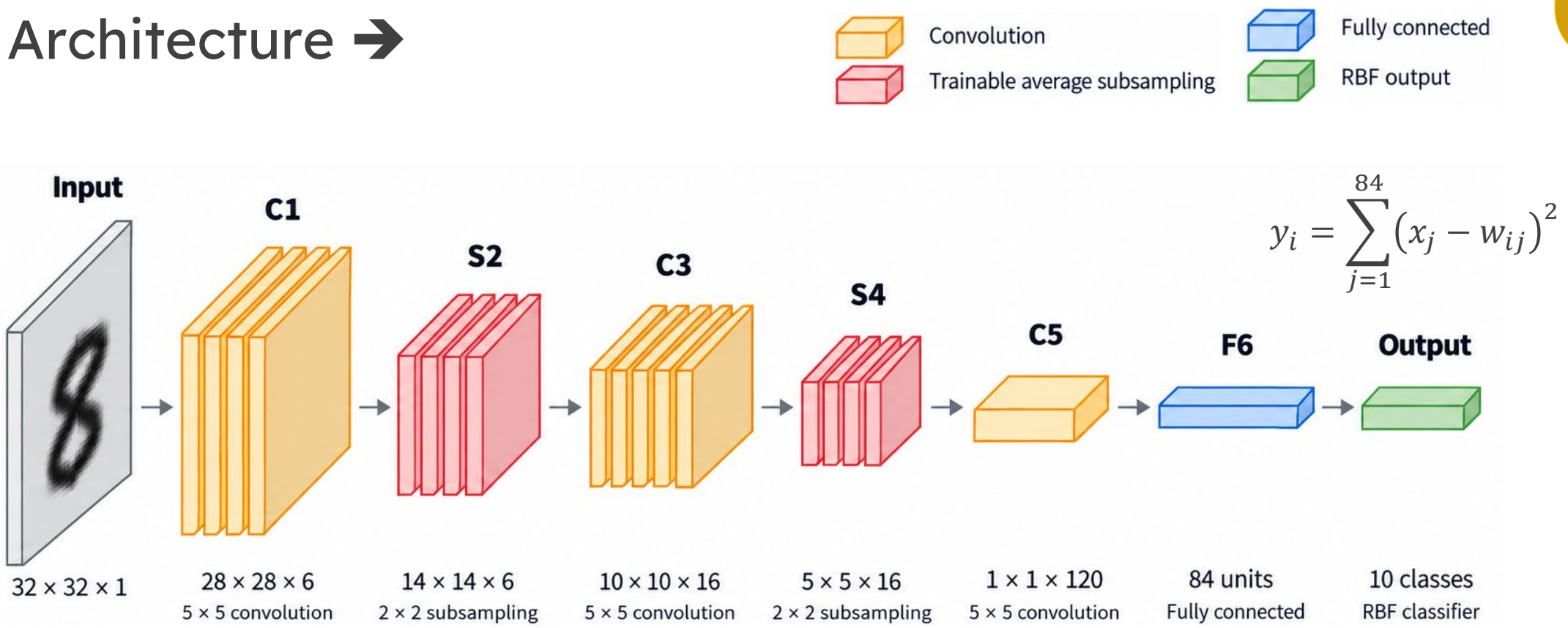
Architecture

Stage	Operation	Output Size
Input	**Grayscale Image**	$32 \times 32 \times 1$
C1	5×5 Convolution | 6 Features	$28 \times 28 \times 6$
S2	Subsampling	$14 \times 14 \times 6$
C3	5×5 Convolution | 16 Features	$10 \times 10 \times 16$
S4	Subsampling	$5 \times 5 \times 16$
C5	5×5 Convolution | 120 Features	$1 \times 1 \times 120$
F6	Fully Connected Network	84
Output	**10 number classes**	**10**

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278-2324.

LeNet

Architecture ➔



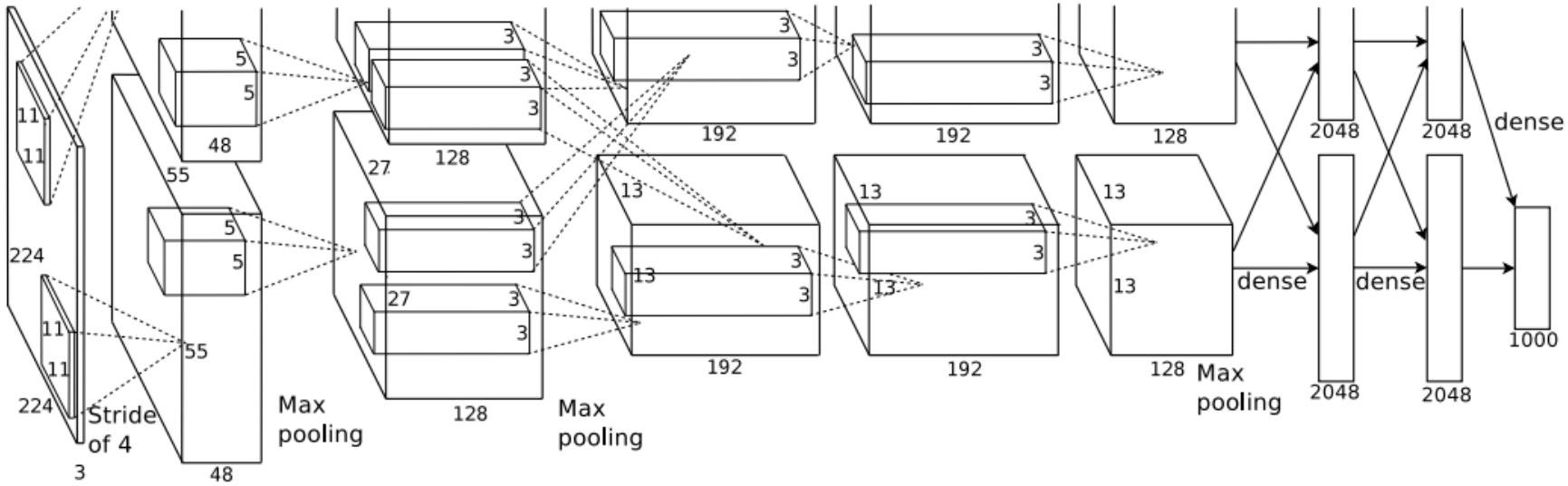
LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278-2324.

AlexNet

AlexNet achieved a breakthrough in the 2012 ImageNet competition. Compared with the small grayscale digits processed by LeNet, ImageNet contains a vast collection of color photographs with much greater variation in object pose, background, lighting, and scale. It consists of five convolutional layers and three fully connected layers, with approximately 60 million parameters. Its success came from a combination of model capacity, GPU computing, ReLU activations, data augmentation, and dropout.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems, 25*.
Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. Communications of the ACM, 60(6), 84-90.

AlexNet



The subscripts indicate the number of output channels in each convolutional layer. The final three numbers represent the sizes of the fully connected layers, with 1,000 outputs corresponding to the ImageNet classification categories.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems, 25*.
Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. Communications of the ACM, 60(6), 84-90.

AlexNet

ReLU: Supporting the training of deep networks

Uses ($\mathrm{ReLU}(x) = \max(0, x)$) to support efficient training by avoiding saturation for positive inputs.

GPU Computing: Making large-scale convolutional training practical

Accelerates convolution through parallel processing. AlexNet used two GPUs to train its large network.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems, 25*.
Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. Communications of the ACM, 60(6), 84-90.

AlexNet

Dropout: Reducing excessive dependence on particular neurons

Randomly disables some neuron outputs during training, reducing reliance on specific features and helping prevent overfitting.

Data Augmentation: Exposing the model to more image variations

Uses random crops, horizontal flips, and color variations to increase training diversity and improve generalization.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, *25*.
Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. Communications of the ACM, 60(6), 84-90.

VGG

VGG follows a consistent design: it repeatedly uses small convolutional filters while progressively increasing network depth. The widely used **VGG-16** contains **13 convolutional layers and 3 fully connected layers**. The “16” counts these layers with weights, excluding pooling layers and activation functions.

Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.

VGG

ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 **LRN**	conv3-64 **conv3-64**	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 **conv3-128**	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 **conv1-256**	conv3-256 conv3-256 **conv3-256**	conv3-256 conv3-256 **conv3-256**
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 **conv1-512**	conv3-512 conv3-512 **conv3-512**	conv3-512 conv3-512 **conv3-512**
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 **conv1-512**	conv3-512 conv3-512 **conv3-512**	conv3-512 conv3-512 **conv3-512**
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

Table 2: **Number of parameters** (in millions).

Network	A,A-LRN	B	C	D	E
Number of parameters	133	133	134	138	144

With stride 1, no dilation, and no downsampling between layers:

- Two stacked 3 × 3 convolutions have a theoretical receptive field of **5 × 5**.
- Three stacked 3 × 3 convolutions have a theoretical receptive field of **7 × 7**.

The receptive field is the region of the input that can influence a particular output location.

Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.

VGG

Stacking two smaller convolutions uses fewer parameters and allows an additional nonlinear transformation between them. However, having the same receptive field does not make the operations equivalent. In particular, adding ReLU between the layers changes the function they represent.

Although its convolutional blocks use small filters, the complete VGG-16 still has approximately 138 million parameters, with a large proportion concentrated in the fully connected classifier.

Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.

RasNet

Increasing network depth does not necessarily improve performance. The ResNet paper observed that some deeper plain networks had even higher training error, a phenomenon known as the degradation problem.

This differs from typical overfitting. In overfitting, a model performs well on the training set but poorly on the validation set. With degradation, the deeper model fails to achieve better performance even on the training data.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

RasNet

The Core Idea of Residual Learning

Suppose the desired mapping for a stack of layers is ($H(x)$). ResNet reformulates it as:

$$H(x) = x + F(x)$$

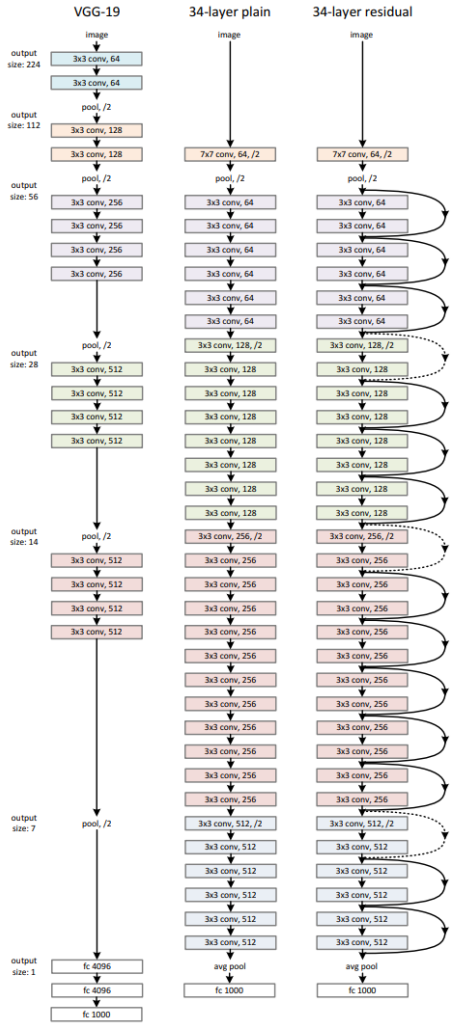
The parameterized branch therefore learns the residual:

$$F(x) = H(x) - x$$

Meanwhile, the input (x) is added directly to the branch's output through a shortcut connection.

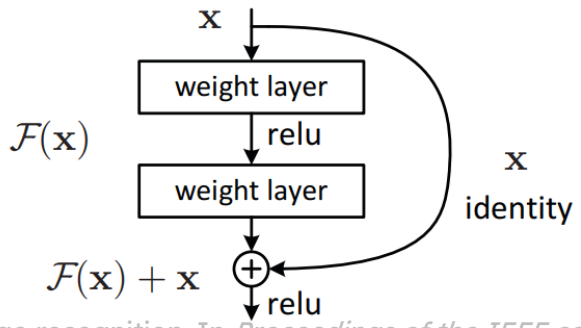
He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

RasNet



For example, if the existing features are already useful, the next block may only need to preserve them. In this case, the residual branch can learn a correction close to zero, rather than reconstructing the entire identity mapping through multiple nonlinear layers.

$$y = ReLU(F(x) + x)$$



He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

RasNet

Why Do Shortcuts Help Gradient Flow?

Consider the result of the addition, before applying ReLU:

$$z = x + F(x)$$

$$\frac{\partial z}{\partial x} = I + \frac{\partial F(x)}{\partial x}$$

Here, (I) is the identity matrix contributed by the identity shortcut. It provides a gradient path that bypasses the residual branch. This helps optimization, but it does not guarantee that networks of arbitrary depth can always be trained successfully.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

RasNet

The first (1×1) convolution in a bottleneck block typically reduces the number of channels, allowing the middle (3×3) convolution to operate on a narrower representation. The final (1×1) convolution then increases the number of channels.

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

RasNet

Although a (1×1) convolution does not expand the spatial receptive field, it still combines information across channels.

The two tensors being added must have compatible shapes. When the spatial dimensions or channel count change, a (1×1) projection shortcut with an appropriate stride is commonly used.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

RasNet

ResNet typically uses global average pooling before its final classification layer. This averages the spatial values within each channel, producing a feature vector that is passed to the classifier.

Therefore, greater depth does not necessarily mean more parameters. A standard ResNet-50 has approximately 25.6 million parameters, fewer than the complete VGG-16 model.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

Data Augmentation

Data augmentation

It applies transformations to training samples, exposing the model to a wider range of possible inputs.

For example, the same cat may appear on the left or right side of an image, and the image may be slightly brighter or darker depending on the lighting conditions. We want the model to recognize the cat despite these variations.

Data Augmentation

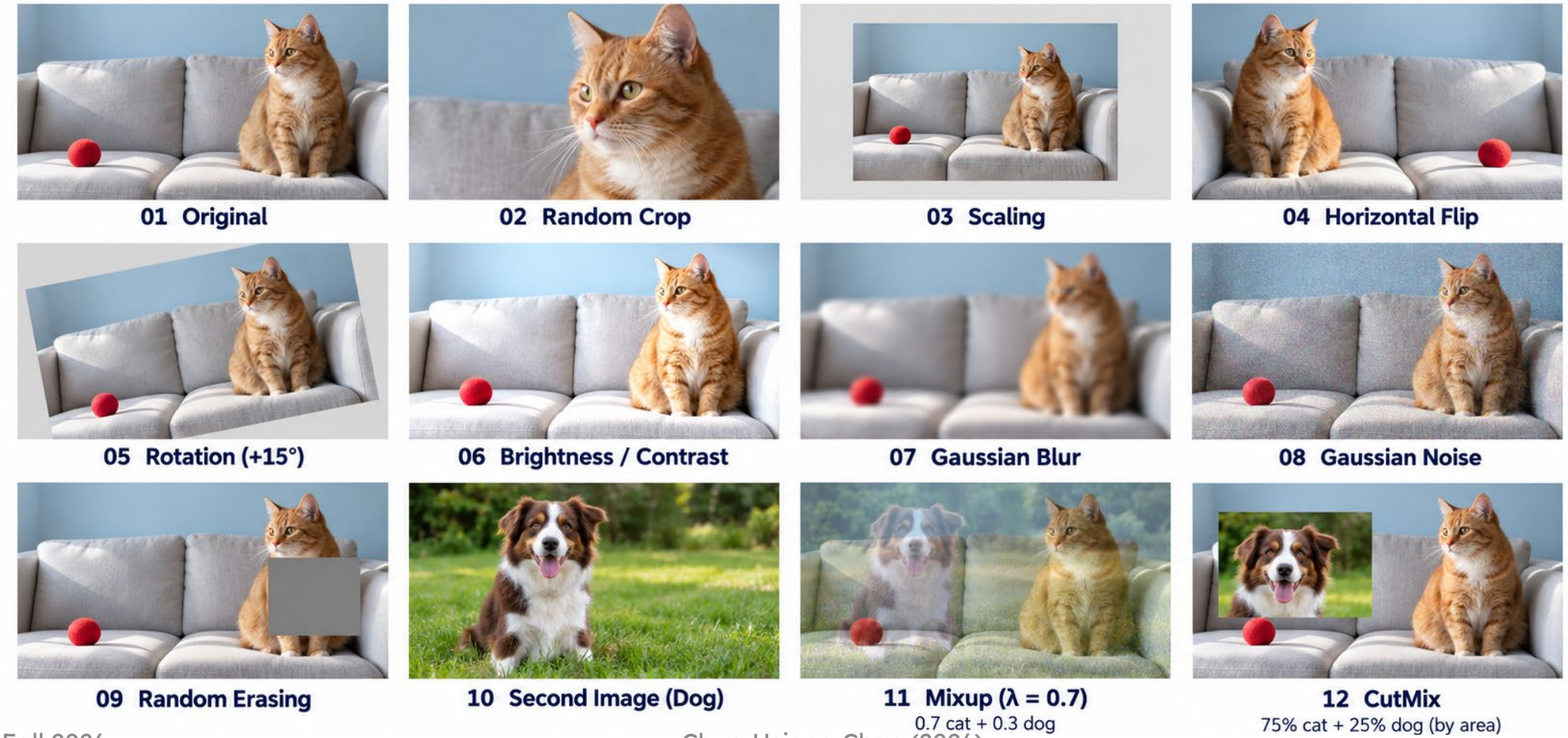
For standard label-preserving augmentation:

$$(x, y) = (T(x), y)$$

Here, (T) is a random transformation.

	Intended Variation	Key Considerations
Random cropping and scaling	Position and scale	Avoid frequently cropping out the target entirely.
Horizontal flipping	Left–right orientation	May be unsuitable for text or tasks where left and right have distinct meanings.
Small-angle rotation	Camera tilt	Excessive rotation may change the meaning of digits or symbols.
Brightness and contrast adjustments	Lighting conditions	Preserve important color cues.
Blur and noise	Image quality	Use intensities that reflect realistic conditions.
Random erasing	Partial occlusion	Excessive occlusion may remove the visual evidence supporting the label.

Data Augmentation



Data Augmentation

Mixup: mixed image and label

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j$$
$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

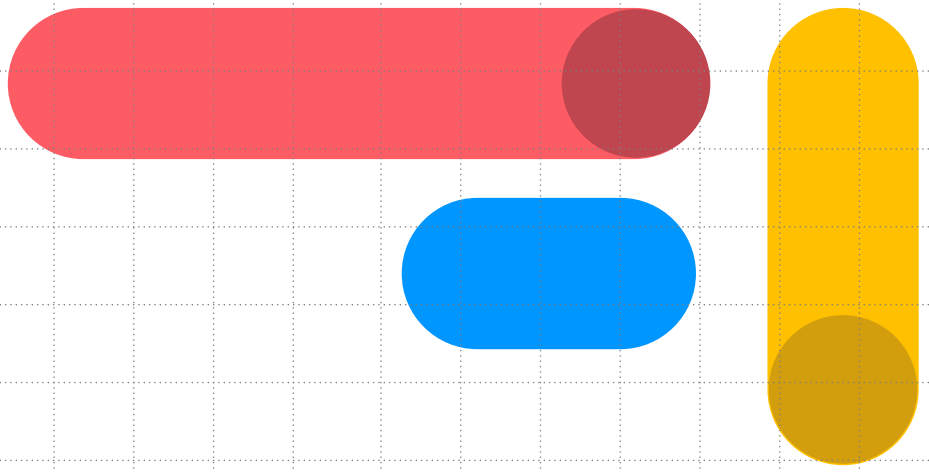
For example, combining a cat image with a weight of 0.7 and a dog image with a weight of 0.3 also produces a soft training label with the same proportions.

This encourages smoother changes in the model's predictions between samples.

Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2017). mixup: Beyond empirical risk minimization. *arXiv preprint arXiv:1710.09412.*

References

1) LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), 2278-2324.
2) Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. Advances in neural information processing systems, 25.
3) Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. Communications of the ACM, 60(6), 84-90.
4) Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.
5) He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).
6) Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2017). mixup: Beyond empirical risk minimization. *arXiv preprint arXiv:1710.09412*.



The End

Thank you for your attention!

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